

Quantum Mechanics III

Lecture 14

Automated Instrument Calibration

- Operational setting
- Mathematical representation
- Dynamics and measurement
- Consistency conditions
- Example

■ Operational setting

Preparation and measurement procedures are distinguished from their matrix representations.

$$p(k|i) = \text{Tr}[E_k \mathcal{E}(\rho_i)]$$

■ Mathematical representation

Positivity, trace, and tensor-product conventions are fixed before the main calculation.

$$J_{ai} = \partial p_a / \partial \theta_i$$

■ Dynamics and measurement

The map from a state to observed probabilities is written explicitly for the experiment under discussion.

$$r_* = \operatorname{argmin}_r(L_R, N_{fail}, d_{QDC})$$

■ Consistency conditions

Complete positivity, normalization, Hermiticity, and appropriate classical limits are checked as needed.

■ Example

The example is worked for a qubit or two-mode system before extending the notation to higher dimension.

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Reading: Instrument Closure Protocol ICP-26; Park, ch. 14; Campus refinement guide